How a building stands up

Every building does two jobs: it **holds things up**, and it **stays standing when pushed sideways**. A structural engineer designs the path every load takes to the ground. That is what KOR is paid for, and it is the whole of this lesson.

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JOB ONE

Holding things up: gravity

Gravity loads come in three kinds, and the engineer adds them up floor by floor, all the way down.

- **Dead load:** the building itself. Concrete is heavy, about 2,400 kg per cubic metre, so in a concrete tower the structure is most of what the structure carries.
- **Live load:** people, furniture, cars. The building code sets it by use; a parkade, a classroom and an apartment are each designed for a different number.
- **Snow:** set by location. It matters most for roofs in the Interior and Alberta.

The **load path** is the route each load takes: slab → beams → columns or walls → foundations → soil. Columns stack straight down where they can.

✓ THE WORD THAT COSTS MONEY: TRANSFER

When a column or wall above can't continue straight down (a tower over an open lobby, or a different grid in the parkade), a very deep beam or a thick **transfer slab** carries it sideways to new supports. Transfers are heavy, slow to build and expensive. Avoiding them is one of the most valuable things a structural engineer does, and the only time to do it cheaply is at concept.

JOB TWO

Staying up when pushed: wind and earthquakes

Wind and earthquakes push a building sideways. In coastal BC, **earthquake usually governs**: the Lower Mainland and Vancouver Island are among the most seismic places in Canada. Wind can govern for a tall, slender tower.

- **The lateral system** is whatever resists the push. In a concrete tower it is the **core**: the concrete walls around the elevators and stairs. In wood frame it is **shear walls**, plywood-sheathed walls tied down to the floor below. In steel it is **braced frames** (diagonal members) or **moment frames** (stiff beam-column joints).
- **Floors are diaphragms:** horizontal plates that collect the sideways force at each level and deliver it to the core or shear walls.
- **Ductility:** buildings are designed to bend and absorb a big earthquake without collapsing, not to come through undamaged. The code's main goal is that people get out alive. For buildings over 30 m in the highest seismic categories, the 2020 National Building Code adds a requirement to limit damage as well.

KOR'S OWN LINE ON WOOD

KOR tells clients that the Building Code prohibits shear-wall transfers in wood-frame buildings over four storeys, so KOR should be brought in at schematic design. Five- and six-storey wood lives or dies on walls that stack.

WHERE IT ENDS

The ground

Every load ends in the soil. The **geotechnical engineer** reports what the ground can carry and how it will behave in an earthquake; the structural engineer designs the foundations to suit, and tells the geotechnical engineer the loads, the settlement the building can tolerate and the seismic requirements.

- **Spread footings:** pads under columns and walls, on good ground.
- **Raft:** one thick slab under the whole building, to spread a heavy tower's weight.
- **Piles:** columns driven or drilled down to firmer ground.
- **Soft or liquefiable soils** (Richmond and much of the Fraser delta) mean ground improvement or piles, and real money. Ask early whether there is a geotechnical report.

WHY THE CLIENT CARES

The decisions clients feel

Clients don't buy calculations. They feel a handful of early decisions in cost, schedule and how many units fit, and the structural engineer drives all of them.

- **Core size and location.** EGBC's guideline for tall concrete buildings says the structural engineer should agree core size, location and openings, structural irregularities and column positions with the architect early.
- **The column grid,** and whether it works for both the suites above and the parking below. If not: transfers.
- **Slab thickness.** It sets floor-to-floor height, which sets how many storeys fit under a height limit. An inch saved per floor adds up over thirty floors.
- **Wood, concrete or hybrid.** Five- and six-storey wood frame, and encapsulated mass timber up to twelve storeys, are now real options in BC. Each changes the structure, the schedule and the cost.

✓ WHAT KOR BUILDS

Mostly concrete residential towers and mid-rise wood or concrete multifamily: One Burrard Place, 54 storeys over 8 parking levels (Reliance, Jim Pattison Group); Harwood, 32 storeys (Intracorp); Mod at 1150 Cook, Victoria, 15 storeys; N3 in Calgary, 17 storeys with no car parking.

VOCABULARY

Words to use

- **Load path:** how a load gets to the ground.
- **Gravity system / lateral system:** what holds the building up / what keeps it standing sideways.
- **Core:** the concrete walls around elevators and stairs; a tower's backbone.
- **Shear wall:** a wall that resists sideways force.
- **Diaphragm:** a floor acting as a horizontal plate.
- **Transfer:** a beam or slab that moves a load sideways because the support below isn't there.
- **PT slab (post-tensioned):** a concrete slab with steel tendons stretched tight after it hardens, so it can be thinner and span further. Standard in Vancouver towers.
- **Drift:** how far a floor moves sideways in wind or an earthquake.
- **SER (Structural Engineer of Record):** the engineer responsible for the structure as a whole.
- **Geotech:** the geotechnical engineer; the soil specialist.

IN THE ROOM

What to ask, and what not to say

Five questions for an architect at concept. Each shows you know where the cost is, and each is a reason to bring KOR in now.

1. Where is the core, and is it fixed yet?
2. Does the tower grid line up with the parking below, or are we looking at transfers?
3. What's your height limit and floor-to-floor? How many storeys are you trying to fit?

4. Is there a geotechnical report yet? Anything soft or wet on the site?
5. Concrete, wood or a hybrid, and has a five- or six-storey wood option been tested?

> ✕ DON'T
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> Don't give sizes, thicknesses or costs, even rough ones. The right answer is "Good question, let me get [engineer] on a call this week." That line also gets you the second meeting.

TWO MINUTES

Check yourself

1. What are the two jobs of a structure?

Carry gravity loads to the ground, and resist sideways loads from wind and earthquakes.

2. In a concrete tower, what resists the earthquake?

Mostly the core: the concrete walls around the elevators and stairs.

3. Why do transfers matter to a developer?

They're heavy, slow and expensive, and they can only be designed out early.

4. Why does slab thickness matter?

It sets floor-to-floor height, so it decides how many storeys fit under a height limit.

5. A site in Richmond. What do you ask first?

Is there a geotechnical report? Soft or liquefiable soil means ground improvement or piles.

> SOURCES, AND THE REVIEW THIS STILL NEEDS
>
> KOR's own portfolio and proposal text; EGBC's Professional Practice Guidelines for tall concrete, Part 3 and geotechnical services; the joint guidelines for encapsulated mass timber. A KOR engineer reads each lesson before it is used with a client.